

# SupportDesk

## Performance Case Study — Enterprise Ticketing Platform

### OVERVIEW

SupportDesk is a React and Next.js ticketing platform for managing customer support tickets: ticket management, search, advanced filtering, agent assignment, and support operations tracking. As ticket volume grew to 10,000 records, the Tickets page began showing significant rendering and interaction delays.

### THE PROBLEM

Users reported a slow initial page load, laggy search, and slow filtering. An initial React Profiler recording measured a render time of 3675ms.

### ROOT CAUSE ANALYSIS

- The full 10,000-ticket dataset was regenerated on every render instead of once.
- Each filter change re-filtered, re-sorted, and re-rendered the entire ticket table.
- Every search keystroke triggered a full dataset scan, sort, and re-render, with no memoization and additional unnecessary computation in the search path.

### OPTIMIZATIONS IMPLEMENTED

- Memoized dataset generation so the 10,000-ticket dataset is created once and reused.
- Memoized filtering logic so filters recompute only when their inputs change.
- Added debounced search, `useDeferredValue`, and precomputed search text.
- Removed unnecessary computation from the search path.
- Paginated the ticket table to 50 rows per page instead of rendering all 10,000 at once.

## RESULTS

Render time improved from 3675ms to 2.3ms, validated with React Profiler.

| Metric          | Before | After  |
|-----------------|--------|--------|
| Render Time     | 3675ms | 2.3ms  |
| Search          | Laggy  | Smooth |
| Filter Response | Slow   | Fast   |
| Rows Rendered   | 10,000 | 50     |
| CPU Usage       | High   | Lower  |
| Responsiveness  | Poor   | Strong |

## TECHNIQUES USED

React Profiler, useMemo, useDeferredValue, debounced search, memoized computation, pagination, large-dataset optimization.

## LINKS

Before / after optimization recordings and this document are linked from the SupportDesk case study on the portfolio site.

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